PROJECT REPORT
Insurance Project

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Batch/Programme: 2023-2027 | B.Tech - Information Technology
College : Kalinga Institute of Industrial Technology, Bhubaneswar (KIIT-DU)

Tool Used: Microsoft Power BI

Topic: Insurance Data Analysis Dashboard

Purpose: Identify high-risk policyholders, track claims trends, and analyze policy adjustments.

1. Problem Statement

Large amounts of consumer and claim data are produced by insurance firms. Extracting insights becomes challenging in the absence of appropriate visualisation. The goal of this project is to create an interactive Power BI dashboard for the analysis of consumer behaviour, insurance performance, and claim patterns.

2. Solution & Features

The project includes an interactive dashboard with multiple pages:
- Overview: Key KPIs like Total Premium, Total Claims, Claim Ratio
- Customer Analysis: Age, gender, and regional distribution
- Claims Analysis: Claim amount, status, and policy-wise breakdown
- Risk Analysis: Identification of high-risk customers and policies

3. Dataset Description

Customer demographics, policy details, premium amounts, and claim information are among the insurance-related data included in the dataset. For analytical purposes, it is a synthetic dataset.

4. Technology Stack

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- CSV Dataset

5. **Key Measures (DAX)**

- Total Claims (SUM(Claims_Frequency))
- Average Claims per Policyholder (AVERAGE(Claims_Frequency))
- Claim Severity Distribution (Low/Medium/High)
- Claims Adjustment Impact (SUM(Claims_Adjustment))
- Claim Ratio = Total Claims / Total Premium

Average Claim Amount = DIVIDE(SUM('synthetic_insurance_data'[Claims_Adjustment]), SUM('synthetic_insurance_data'[Claims_Frequency]), 0)

- **Scatter Plot: Premium Amount vs. Claims Frequency**

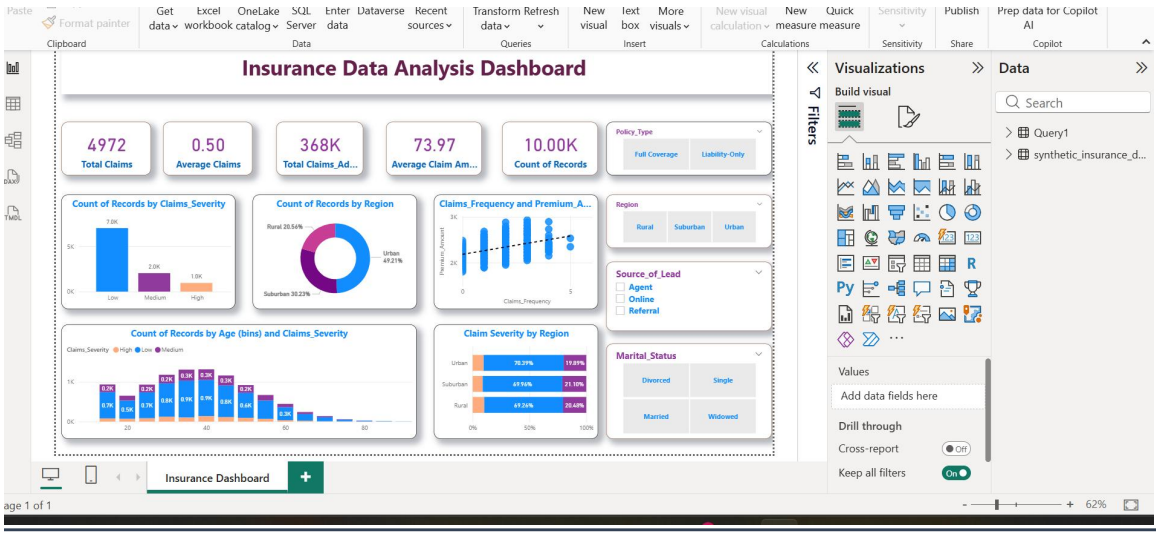
6. Key Insights

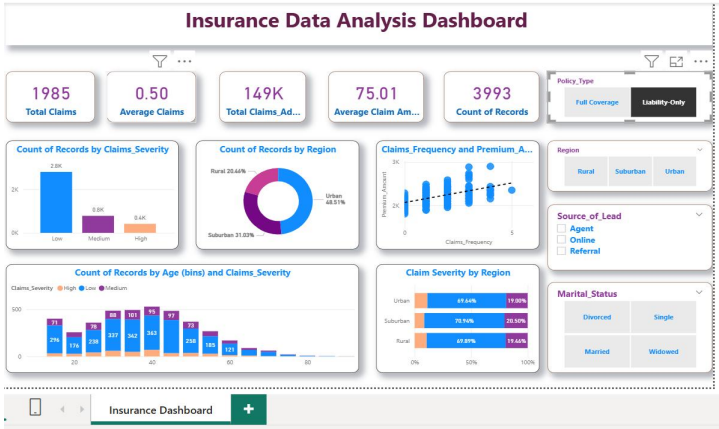
Some areas make more money from premiums than others. Higher claim ratios are found in some policy types.
Demographics of customers affect how they make claims.
Risk locations are indicated by a high claim ratio.

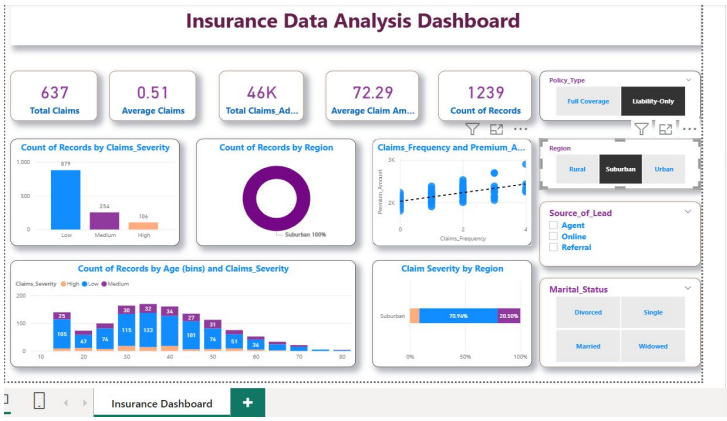
Visuals:

- **Bar Chart: Claims Frequency by Age Group or Region**
- **Heatmap: Claim Severity by Region**
- **KPI Cards: Total Claims, Average Claim Amount**

RESULTS :




Insurance Data Analysis Dashboard
1985
Total Claims
0.50
Average Claims
149K
Total Claims_Ad...
75.01
Average Claim Am...
3993
Count of Records
Policy_Type
Full Coverage
Liability-Only
Count of Records by Claims_Severity
Count of Records by Region
Claims_Frequency and Premium_A...
Region
Rural
Suburban
Urban
Source_of_Lead
Agent
Online
Referral
Count of Records by Age (bins) and Claims_Severity
Claim Severity by Region
Marital_Status
Divorced
Single
Married
Widowed
Insurance Dashboard


Insurance Data Analysis Dashboard
637
Total Claims
0.51
Average Claims
46K
Total Claims_Ad...
72.29
Average Claim Am...
1239
Count of Records
Policy_Type
Full Coverage
Liability-Only
Count of Records by Claims_Severity
Count of Records by Region
Claims_Frequency and Premium_A...
Region
Rural
Suburban
Urban
Source_of_Lead
Agent
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Count of Records by Age (bins) and Claims_Severity
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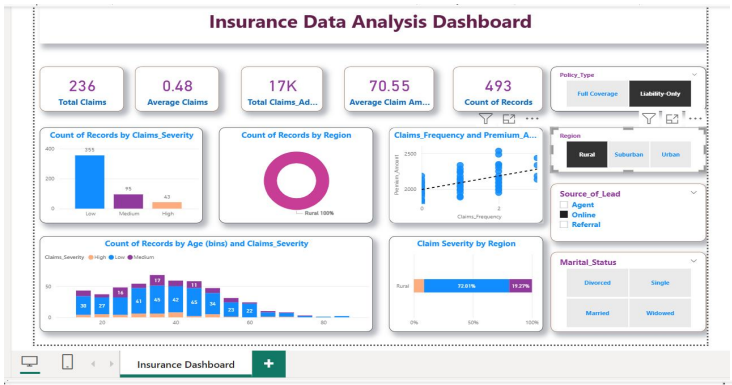
Insurance Data Analysis Dashboard
236
Total Claims
0.48
Average Claims
17K
Total Claims_Ad...
70.55
Average Claim Am...
493
Count of Records
Policy_Type
Full Coverage
Liability-Only
Count of Records by Claims_Severity
Count of Records by Region
Claims_Frequency and Premium_A...
Region
Rural
Suburban
Urban
Source_of_Lead
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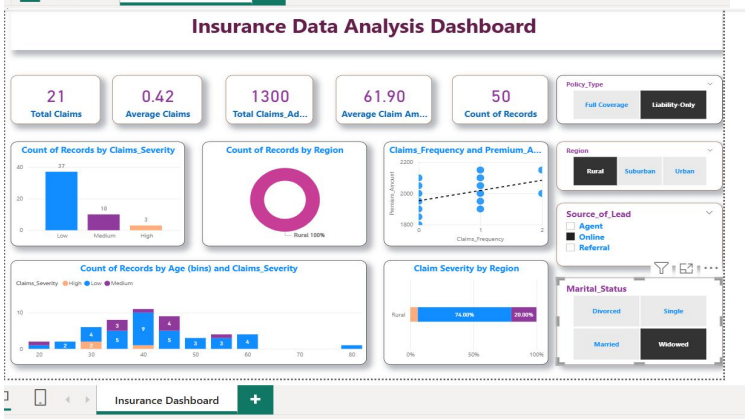
Insurance Data Analysis Dashboard
21
Total Claims
0.42
Average Claims
1300
Total Claims_Ad...
61.90
Average Claim Am...
50
Count of Records
Policy_Type
Full Coverage
Liability-Only
Count of Records by Claims_Severity
Count of Records by Region
Claims_Frequency and Premium_A...
Region
Rural
Suburban
Urban
Source_of_Lead
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7. Unique Points

- Interactive dashboard with filters
- Multi-page analysis
- Business-focused insights
- Clean and simple design

8. Future Improvements

- Add predictive analytics
- Use real-time data
- Integrate machine learning models

9. Conclusion

This project serves as an example of how Power BI can turn unprocessed insurance data into insightful knowledge. It facilitates improved performance, risk analysis, and decision-making.

BY:

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